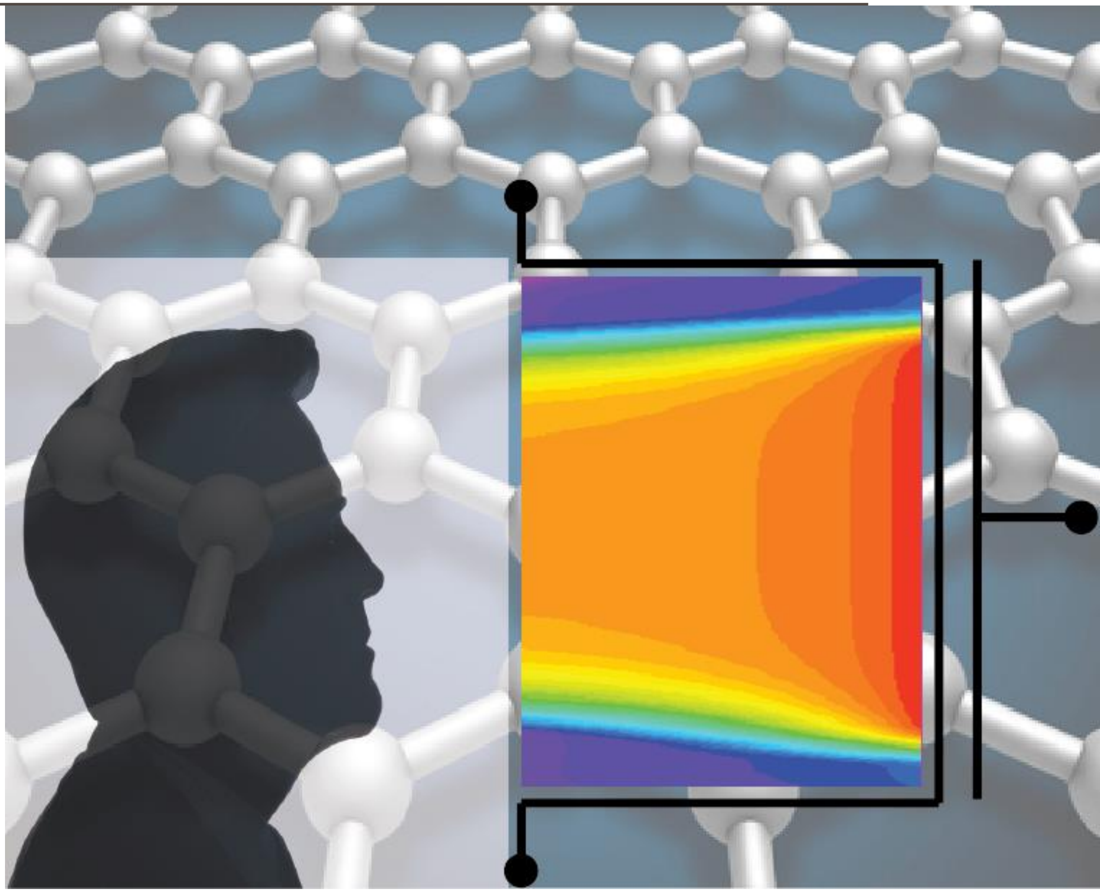
ADVANCED SOLID-STATE DEVICES

Week-2, Lecture-1



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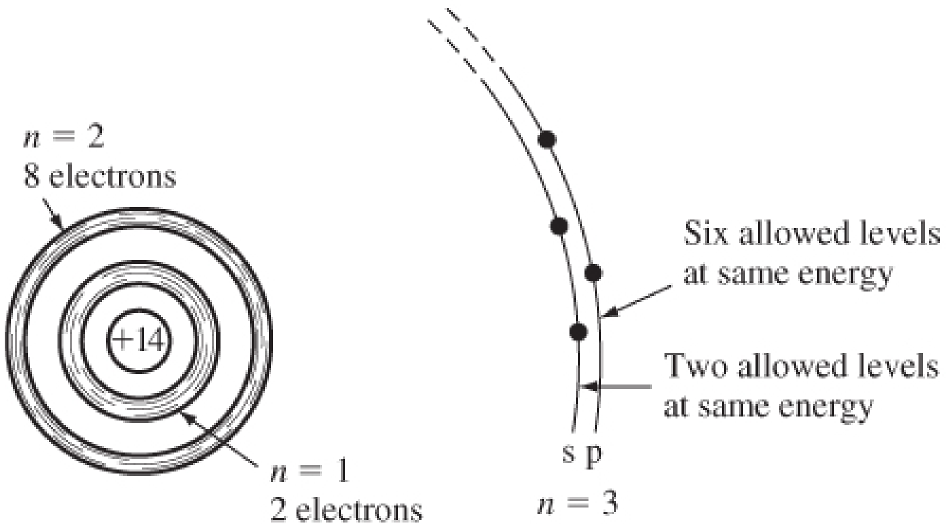
13th January, 2025

Semiconductor Physics

Energy Bands

Formation of Energy Band (4)



(a)

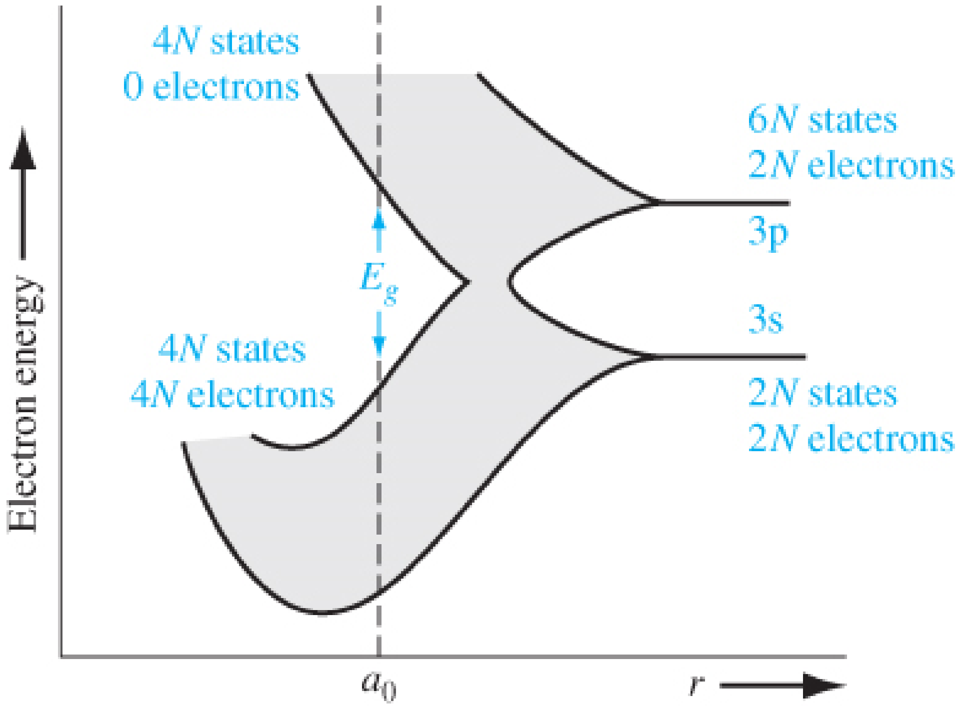
For silicon

- Actual formation of bands is more complicated and governed by Quantum Mechanics
- For silicon, the inner 10 electrons ($n = 1$ and $n = 2$) are tightly bound to nucleus and do not participate in chemical reaction or conduction

Outer Shell

- The outer shell ($n = 3$) has two electrons in 3s and two electrons in 3p (they interact when interatomic distance is reduced)
- The 3p orbital has six allowed levels
- Four levels of 3p are left unoccupied at equilibrium

Formation of Energy Band (5)



For silicon

- When interatomic distance reduces, the 3s and 3p orbitals interact and split
- At equilibrium, four quantum states per atom move in upper energy band and the rest four into lower energy band

Bands and Bandgap

- The lower band is known as ***valence band***, and the upper band is known as ***conduction band***
- The valence band and conduction band are separated by ***bandgap*** $\boldsymbol{E_g}$
- At T=0, all states in valence band will be occupied, while all states in the conduction band will be empty

Semiconductor Physics

Energy Bands and Conduction

Free electrons

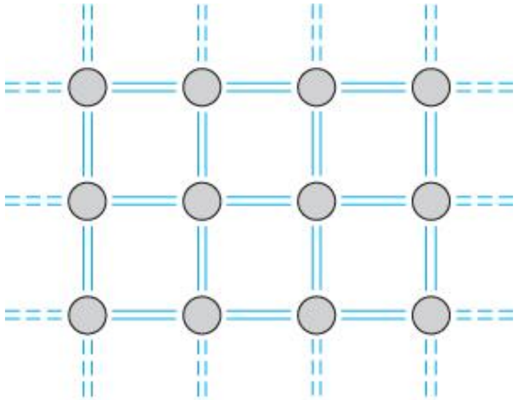


Figure 3.12 | Two-dimensional representation of the covalent bonding in a semiconductor at $T = 0$ K.

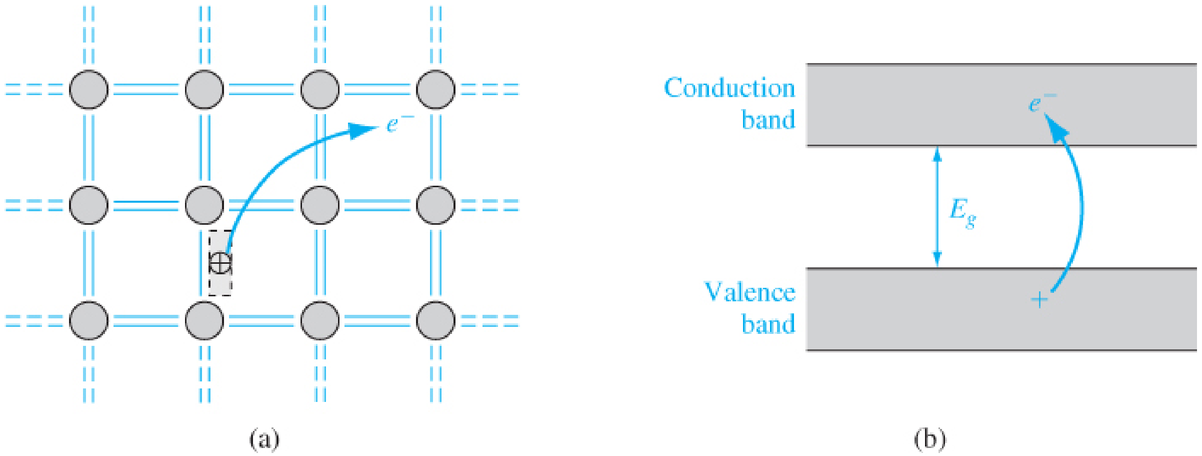


Figure 3.13 | (a) Two-dimensional representation of the breaking of a covalent bond. (b) Corresponding line representation of the energy band and the generation of a negative and positive charge with the breaking of a covalent bond.

For silicon at T=0 K

- Covalent bond: 8 valence electrons
- Valence band filled, conduction band empty

For silicon at T > 0 K

- Covalent bond breaks for electron
- Electrons jump to conduction band and positive charged empty state created in the valence band
- Effect more pronounced as T increased

Holes

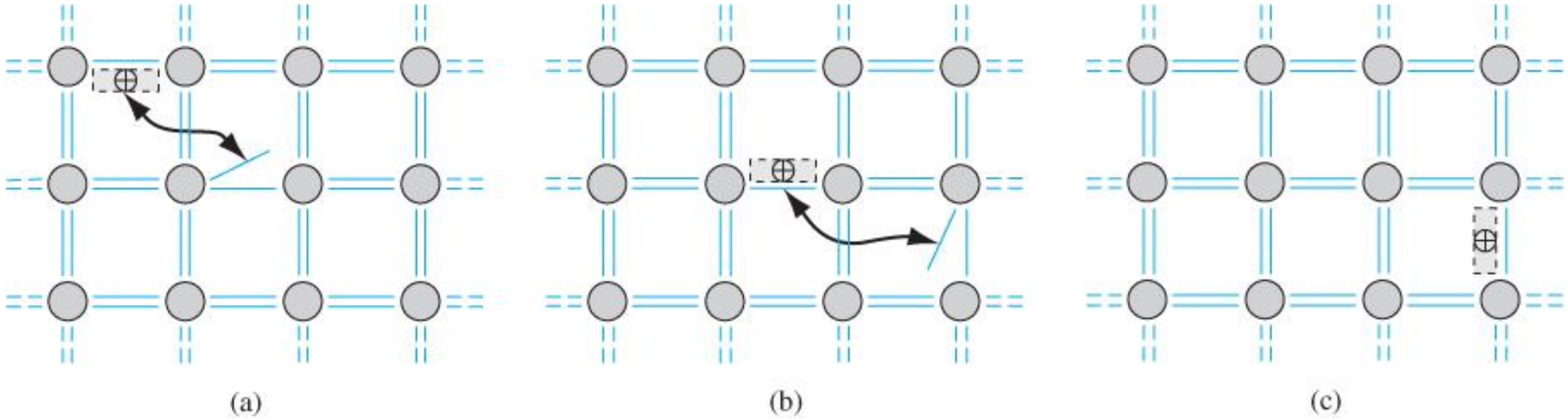


Figure 3.17 | Visualization of the movement of a hole in a semiconductor.

<u>Movement of vacancy</u>

- If there is a vacancy in the valence band, electrons in the valence band can fill up that vacancy in the valence band very easily
- We can consider this movement as movement of “positive charged vacancy”
- This vacancy is called “hole”

Behaviour of an Electron in a Crystal (1)

- ***Drift Current***: current or movement of charge carriers due to the applied electric field ($\vec{E}$)
  - Force on electron $\vec{F} = -q\vec{E}$
- Based on ***Newtonian Mechanics***

- **Differences:**
  1. Will move only if the higher energy levels are **available**
  2. **Interactions** of electron within the crystal
- Actual description requires considering Quantum Mechanics
- We will consider a mix of Quantum Mechanics and Newtonian Mechanics to understand the SSD (to ease our analysis).

Behaviour of an Electron in a Crystal (2)

- The dynamics of electron inside a crystal depends on the crystal structure
- The electron behavior due to force in free space will be different from electron behavior inside the crystal

- Internal forces due to positive charged ions will influence the motion

$$F_{total} = F_{ext} + F_{int} = ma$$

- Accounting for F_{int} is difficult (requires considering Quantum Mechanics)
- Instead, we write:

$$F_{total} = m^* a$$

where m^* is the effective mass

- Intuitively, effective mass is a concept that allows us to apply Newtonian Mechanics for electron motion by approximating the effect of Quantum Mechanics

Free electrons

Problem: Assume that a free electron of mass m has an energy E.

a. What will be its momentum?
b. What will be its wavelength?
c. What will be its linear wavenumber?
d. What will be its angular wavenumber?

Solution: a) $p = \sqrt{2mE}$

b) De Broglie's formula: $\lambda p = h$

$$\Rightarrow \lambda = h/\sqrt{2mE}$$

c) Wavenumber= $1/\lambda = \sqrt{2mE}/h$

d) Angular Wavenumber= $k = \frac{2\pi}{\lambda} = \frac{2\pi\sqrt{2mE}}{h} = \frac{\sqrt{2mE}}{\hbar}$

$$E = \frac{\hbar^2 k^2}{2m}$$

- The E-k relation is known as dispersion relation

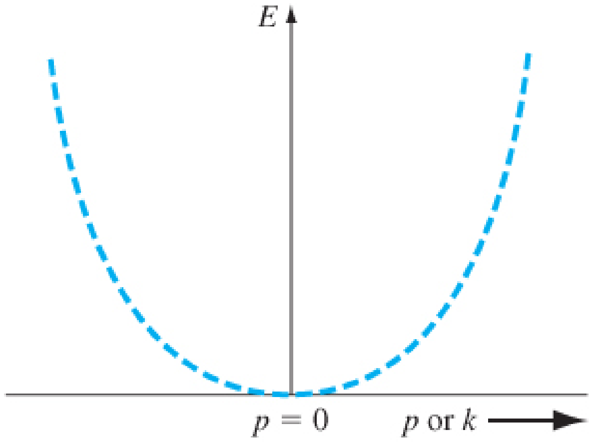


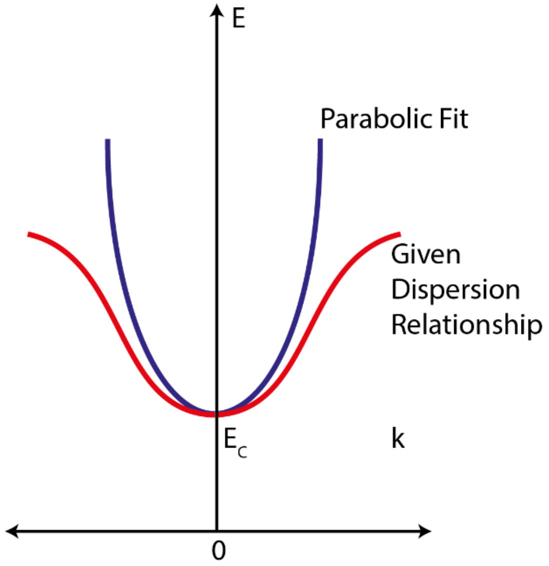
Figure 3.7 | The parabolic E versus k curve for the free electron.

Dispersion Relationship

- The E-k relationship (dispersion relation) in a crystal for an electron may not be parabolic
- We need to compute the dispersion relation with the help of quantum mechanical analysis (it will depend on the crystal structure and will be different for different materials)

- The electronic property of a crystal depends heavily on the dispersion relationship
- We can capture the effect of dispersion relationship using effective mass

Effective Mass Approximation



- Actual Dispersion Relationship can be complex
- Taylor series expansion of the $E-k$ relationship at minima $k=0$

$$E(k) = E(0) + k\left.\frac{dE}{dk}\right]_{k=0} + \frac{1}{2}k^2\left.\frac{d^2E}{dk^2}\right]_{k=0} + \cdots$$

- Parabolic Approximation of Dispersion relationship is done at the minima of the dispersion curve:

$$E(k) = E_C + \frac{\hbar^2 k^2}{2m^*}$$

- Comparing above two equations:

$$\frac{1}{2}k^2\left.\frac{d^2E}{dk^2}\right]_{k=0} = \frac{\hbar^2 k^2}{2m^*} \qquad \Rightarrow m^* = \frac{\hbar^2}{\left.\frac{d^2E}{dk^2}\right]_{k=0}}$$

- We compute effective mass as $m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$ at the minima of the E-k curve.

Effective Mass of a free electron

Problem: What is an effective mass of a free electron?

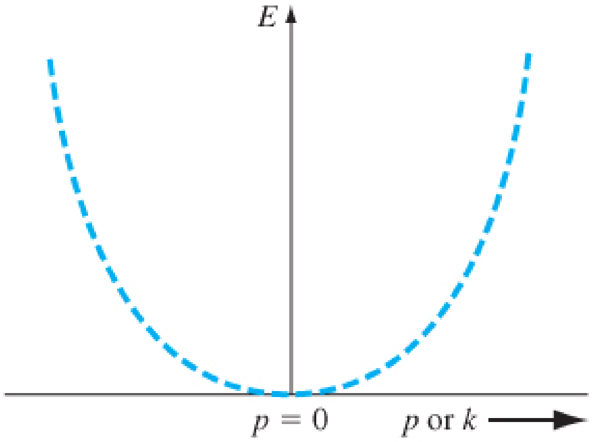


Figure 3.7 | The parabolic E versus k curve for the free electron.

- The E-k relation for a free electron is:

$$E = \frac{\hbar^2 k^2}{2m}$$

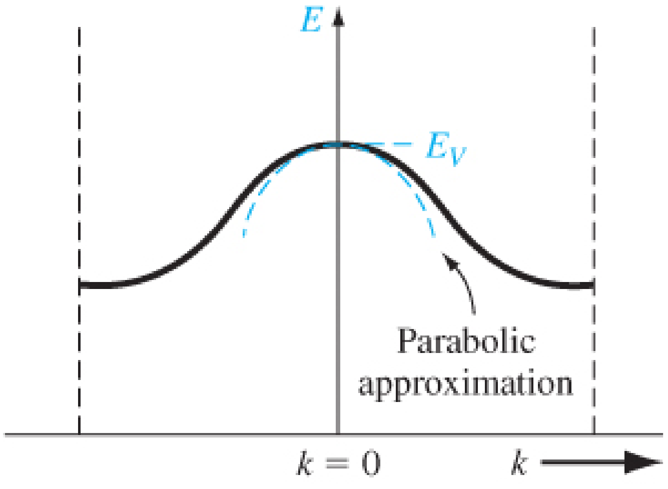
Solution:

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m} \quad \text{and} \quad \frac{d^2E}{dk^2} = \frac{\hbar^2}{m}$$

Effective mass $m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)} = \frac{\hbar^2}{\left(\frac{\hbar^2}{m}\right)} = m$

- The effective mass is same as the rest mass for a free electron

Effective Mass of Electrons in Valence Band



(b)

Problem: The E-k relationship for electrons in the valence band can be approximated as an inverted parabola:

$E(k) = E_V - Ck^2$ [where C is a positive constant]

Compute the effective mass of an electron at the top of the valence band.

Solution:

$\frac{dE}{dk} = -2Ck$ and $\frac{d^2E}{dk^2} = -2C$

Effective mass $m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)} = -\frac{\hbar^2}{2C}$

- The effective mass of electrons at the top of the valence band is negative.

Electrons and Holes

Electrons at the bottom of conduction band:

- Force on electron is in oppositie direction to the electric field (because of negative charge):
$$\vec{F} = -q\vec{E}$$
- Also: $F_{total} = m^* a$
- Acceleration in an electric field: $\vec{a} = \frac{-q\vec{E}}{m^*}$

- Electron at the bottom of the conduction band accelerates in direction opposite to the Electric Field (effective mass m^* is positive)

Electrons at the top of the valence band:

- Force on electron is in oppositie direction to the electric field (same as in conduction band):
$$\vec{F} = -q\vec{E}$$
- Also: $F_{total} = m^* a$

- But in this case effective mass m^* is negative
- Acceleration in an electric field: $\vec{a} = \frac{-q\vec{E}}{-|m^*|} = \frac{q\vec{E}}{|m^*|}$
- Electron at the top of the valence band accelerates in the same direction as the Electric Field

- Instead of electron we can consider hole with a positive charge and positive effective mass
- Concept of hole helps get rid of negative effective mass

Effect of Direction on current Flow

- Interatomic distance dependent on the direction through the crystal

- Electrons travelling in different directions experience different kinds of potential function

- In general, E-k diagram (dispersion relation) depends on the direction in the crystal

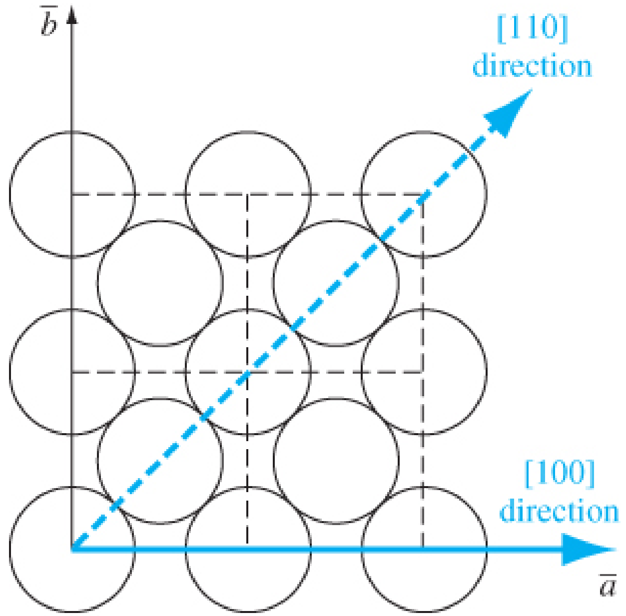


Figure 3.24 | The (100) plane of a face-centered cubic crystal showing the [100] and [110] directions.

E-k diagram of GaAs

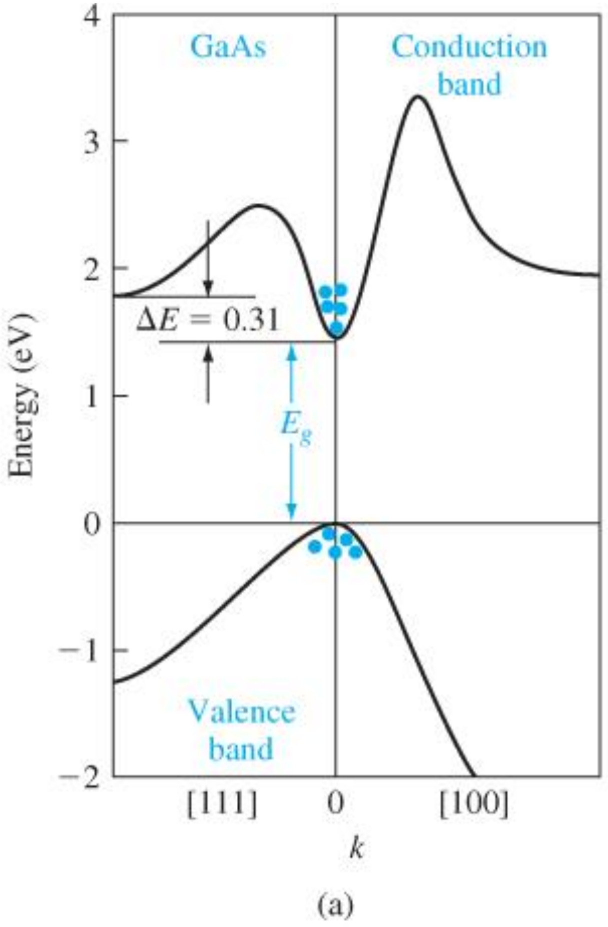
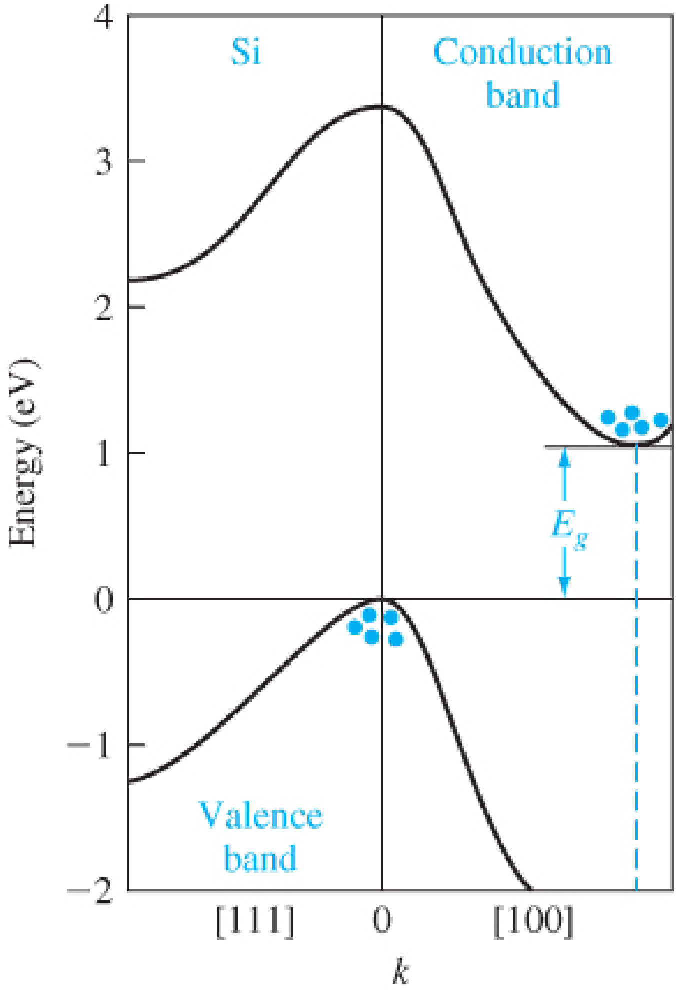


Figure 3.25 | Energy-band structures of (a) (
(From Sze [12].)

- Plot along two directions of the crystal: [100] and [111]
- The minimum energy for the conduction band and maximum energy for the valence band occurs at $k = 0$
- *Direct bandgap semiconductor*

- The electrons will tend to occupy low energy states
- Electrons tend to be at the bottom of the conduction band
- Holes tend to be at the top of the valence band

E-k diagram of Silicon



For silicon

- The valence band energy maximum occurs at $k = 0$
- The conduction band minimum occurs along [100] direction at some value of k
- *Indirect bandgap semiconductor*

Direct Bandgap and Indirect Bandgap (1)

- When an electron transitions from valence band to conduction band or vice versa, following quantities needs to be conserved:
  - Energy
  - Momentum
- Momentum and k are linearly related ($p = \hbar k$)

Electron transition between bands:

➢ Direct bandgap semiconductor requires **only change in energy**

➢ Indirect bandgap material requires change in **both energy and** $\boldsymbol{k}$ (momentum)

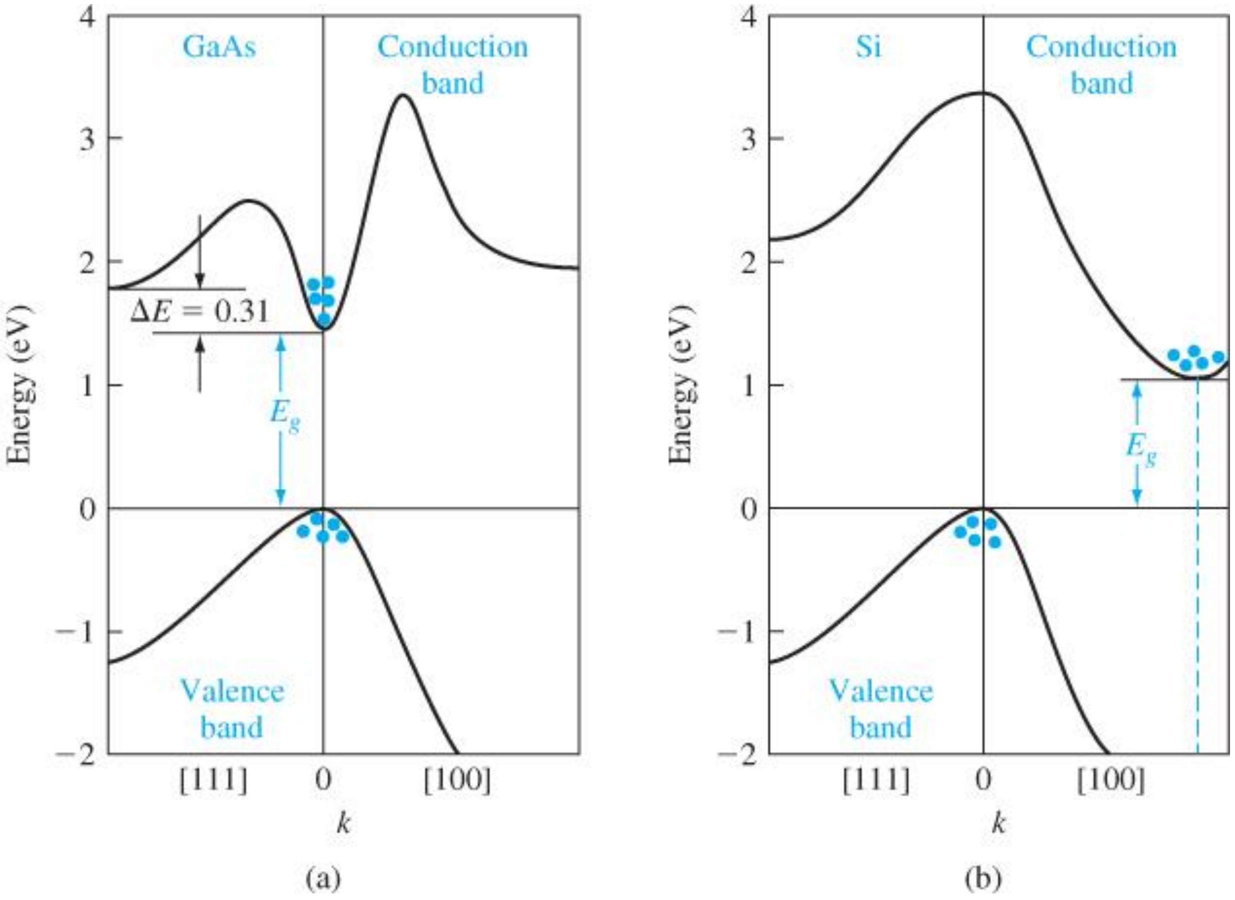
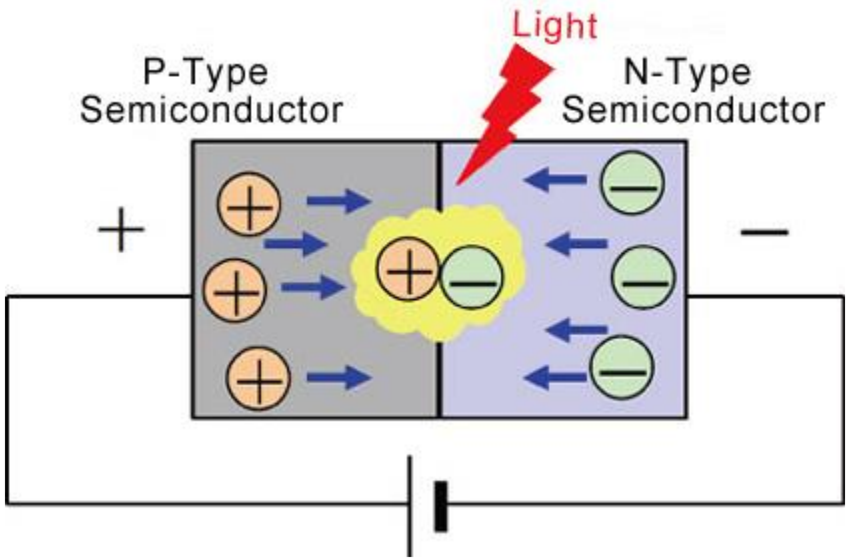


Figure 3.25 | Energy-band structures of (a) GaAs and (b) Si. *(From Sze [12].)*

Direct Bandgap and Indirect Bandgap (2)

Consider Radiative Recombination (example, LEDs)

- Electrons in the conduction band combine with holes in valence band
- The extra energy is released as photon (light)



- Interactions among electrons (holes), photons, phonons, and other particles are required to satisfy conservation of energy and crystal momentum
- Phonons: quantum of vibrational energy that arises from oscillating atoms within a crystal

Direct Bandgap and Indirect Bandgap (3)

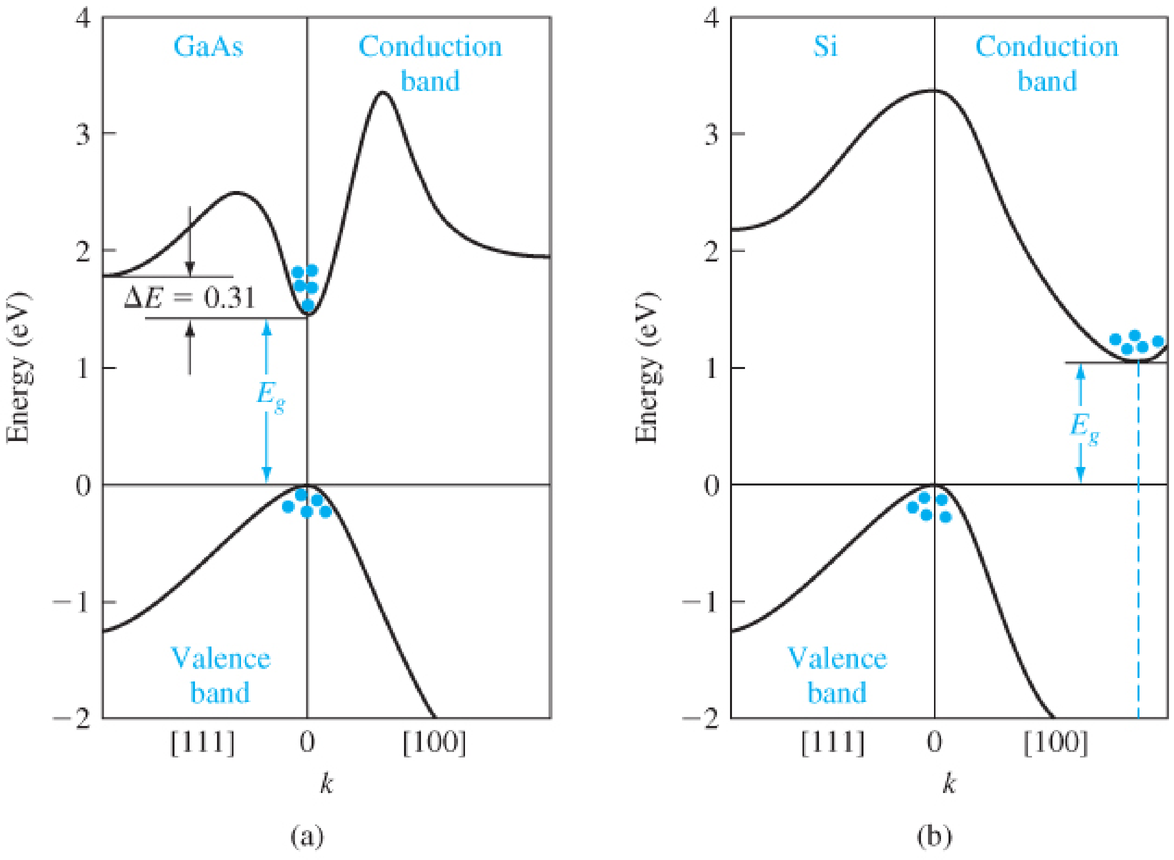


Figure 3.25 | Energy-band structures of (a) GaAs and (b) Si.
(From Sze [12].)

- **Direct bandgap material:** can emit photon without change in momentum (can be easily done)
- **Indirect bandgap material:** requires emission/absorption of the phonon that balances the change in momentum

Direct bandgap material preferred for LED applications

Density of States (DOS)

Density of States (1)

- For a bound electron (as in a crystal) the energy interval is divided into discrete states.
- Within the same interval there can be different number of states at different energies

- We have seen that current flows:
  - When carriers are present
  - When carriers have available states to go
- Density of States (DOS) is a key concept in SSD that determines the presence (concentration) of carriers and availability of states for carriers to go

DOS: Definition (1)

- Density of states (DOS) of a system $D(E)$ describes the number of states available to be occupied per energy interval (between E and $E + dE$) at a given energy level E

- DOS represents if we increase the energy by an amount dE, how many states would be covered:

$$\boldsymbol{D(E)dE}$$

- A high DOS at a specific energy level means that there are many states available for occupation.

- A DOS of zero means that no states can be occupied at that energy level.

- DOS for electrons governs current flow (as we will see later)

DOS: Definition(2)

- DOS represents if we increase the energy by amount dE, how many states would be covered:

$$\boldsymbol{D(E)dE}$$

⇒If we integrate $\boldsymbol{D(E)}$ up to an energy $\boldsymbol{E}$, we should get the number of states less than an energy $\boldsymbol{E}$

$$\boldsymbol{N(E) = \int_{-\infty}^{E} D(E')\, dE'}$$

[$N(E)$ represents the total number of states with energy less than or equal to E]

⇒$\boldsymbol{D(E) = \frac{dN(E)}{dE}}$

- We can use the above relation to compute density of states

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*